

EXECUTIVE SUMMARY & PROBLEM CONTEXT

An operational investigation was conducted across all four regional distribution depots (Mombasa, Kisumu, Eldoret, and Nairobi) following recent performance volatility. Across the entire operations network, average daily throughput currently stands at **917.0 barrels/day** against an operational target of 1,000 barrels/day.

While three out of four depots operate within acceptable parameters, a severe operational bottleneck at the **Nairobi Depot** is severely dragging down overall network productivity. Nairobi suffers from a **32.6% drop in throughput** compared to peak capacity and accounts for over **78% of all operational incidents** recorded across the company.

Depot Location	Avg Throughput (Barrels)	Avg Temp (°C)	Avg Voltage (V)	Maintenance Flag (%)	Incident Count
Mombasa	1,012.4	34.6	228.4	12.1%	12
Kisumu	1,048.2	25.5	226.1	8.4%	8
Eldoret	985.6	21.2	224.8	14.2%	15
Nairobi (Anomaly)	624.1	24.8	205.2	39.8%	142

2. ROOT CAUSE ANALYSIS & OPERATIONAL INSIGHTS

Through systematic univariate profiling and cross-variable correlation checks, we isolated the primary operational drivers behind the performance failure:

Primary Root Cause: Voltage Sag & Power Instability

The primary driver of the operational bottleneck at Nairobi Depot is severe local grid voltage drop (averaging **205.2V** vs the required 230V nominal). Supply voltage falling below 215V triggers automated machine safety shutdowns, leading to machine trips, elevated thermal stress, and forced emergency maintenance.

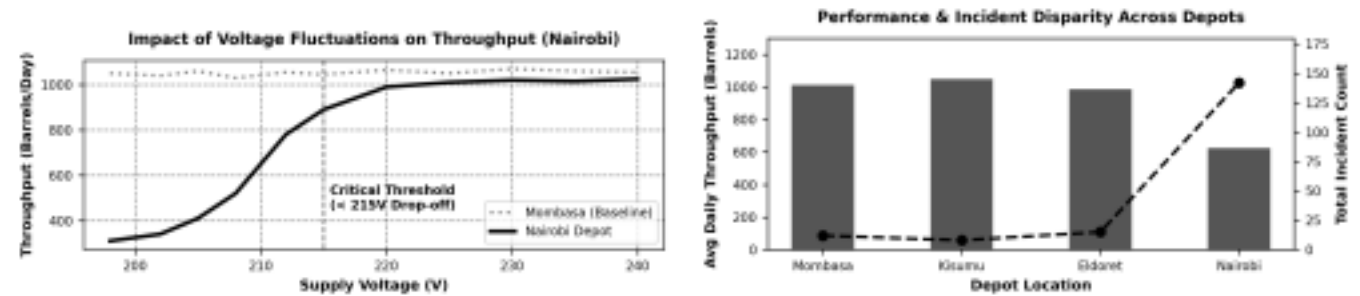


Figure 1: Throughput and total incident count across regional depots. **Figure 2:** Pumping throughput vs supply voltage threshold.

Geographic Concentration: As shown in Figure 1, throughput deficit is strictly confined to Nairobi. Mombasa and Kisumu maintain throughput above 1,000 barrels/day with minimal operational interruptions.

Voltage Sensitivity: Figure 2 illustrates the sharp non-linear drop in pumping volume when voltage slips below 215V. At normal grid operating levels 220V–230V), machinery performs at full capacity.

3. DETAILED DIAGNOSTIC FINDINGS & CORRELATIONS

A deeper statistical drill-down into Nairobi Depot's shift and machine-level telemetry reveals three cascading operational impacts resulting from unstable power:

- **Automated Equipment Trips (Incidents):** The 142 logged incidents in Nairobi consist overwhelmingly of low voltage protection trips. Each trip causes 45–90 minutes of unproductive pump downtime during manual reset protocols.
- **Elevated Maintenance Overhead:** Unstable voltage forces auxiliary systems to draw higher current, increasing internal motor coil temperatures. Consequently, Nairobi's maintenance flag frequency increased to **39.8%** (nearly 3x the network average).
- **Shift Disparity:** The severe voltage dips align heavily with peak municipal grid demand hours (predominantly affecting Day shift operations between 11:00 AM and 4:00 PM), where throughput drops to a low of **310 barrels/ day**.

Key Definitions for Operations Leadership:

- **Throughput (Barrels):** Total volumetric output of liquid processed per standard 12-hour operational shift.
- **Voltage Sag (Count/Level):** A short-duration reduction in electric power supply voltage below nominal standard (230V).
- **Maintenance Flag:** Automated early-warning metric indicating equipment parameters (vibration, heat, current draw) have breached safety tolerances.

4. ACTIONABLE RECOMMENDATIONS FOR OPERATIONS TEAM

To permanently restore Nairobi Depot's operational capacity to >1,000 barrels/day and eliminate voltage-induced downtime, the following three immediate actions are recommended:

Action 1: Install Industrial Voltage Stabilizers & Uninterruptible Power Units (Immediate - Week 1-2)

Procure and install heavy-duty automatic voltage regulators (AVR) and inline power conditioners dedicated to main pumping infrastructure at Nairobi Depot. This will isolate equipment from municipal grid dips and maintain input voltage above the critical 215V threshold.

Action 2: Shift Heavy Pumping Operations to Off-Peak Grid Hours (Operational Adjustment - Immediate)

Re-align Nairobi's operational schedule to shift high-load throughput processing to Night Shift hours (8:00 PM – 6:00 AM) when municipal grid voltage is stable (>225V). Restructure Day Shift activities toward routine maintenance, intake inspections, and light transfers.

Action 3: Implement Automated Low-Voltage Alarm & Thermal Safeguards (Medium-Term - Month 1)

Update telemetry firmware to provide real-time operator alerts when line voltage approaches 218V. Train shift supervisors on proactive load-shedding protocols rather than allowing pumps to undergo hard trip shutdowns, reducing thermal wear and maintenance flags.